TOC CIE-1:

PART A-

Q1. $\delta^*(q_0, 001)$.

$\delta^*(\delta(q_0, 0), 01)$

$\delta^*(q_1, 01)$

$\delta^*(\delta(q_1, 0), 1)$.

$\delta^*((q_1, q_2), 1)$

~~$\delta^*(\delta(q_1, 1)$~~

$\delta(q_1, 1) \cup \delta(q_2, 1)$.

$\{q_2, q_3, q_4\}$.

Q2. ε-closure $(3) = \{3, 6, 7, 8, 10\}$

ε-closure $(5) = \{5, 6, 7, 8, 10\}$

ε-closure $(12) = \{12, 13, 1, 2, 4\}$.

$\therefore$ ε-closure $(3, 5, 12) = \{1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 13\}$.

Q3. i) bbbaa.

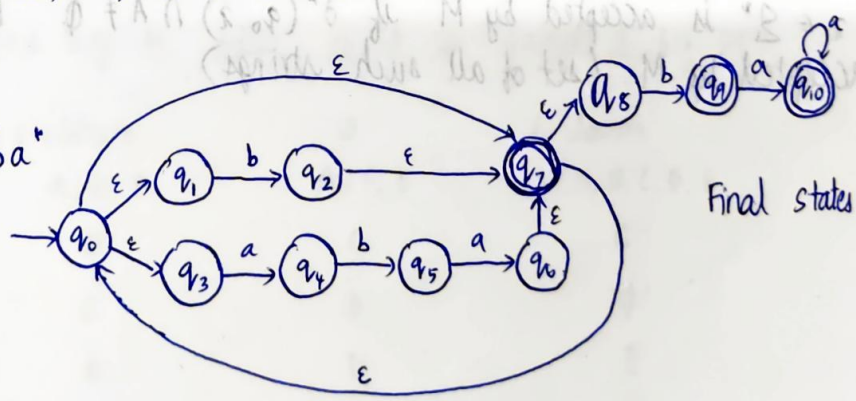
ii) a.

Q4. $L^2 = \{ab, ac, c, abac, abc, acc, acab, cab, cac\}$.

$L^4 = (L^2)^2 = \{ab, ac, c, \ldots\ldots$ 81 elements.

Q5. $(b+aba)^* ba^*$.

For $(b+aba)^* ba^*$



Final states: q_9, q_{10}.

PART-B:

Q1.

a) Regular expression is recursively defined as the formula for expressing regular languages. If Σ is an alphabet, the set R of regular languages if the language ϕ is an element of R and for every $a \in \Sigma$, $\{a\}$ is in R. For any 2 languages L_1, L_2 in R, $L_1 + L_2$, $L_1 L_2$ and L_1^* is in R.

i) $(0+1)^* (00+10+11) + 0 + 1 + \varepsilon$.

ii) $(1+01)^* (0+\varepsilon)$.

Q1.

b) 5-tuple machine: $M = (Q, \Sigma, \delta, q_0, F)$.

Q: set of all states (finite, non-empty).

Σ: alphabet (finite set).

q_0: start states

F: set of final states.

δ: transition function.

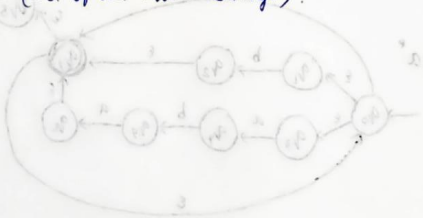
Extended transition function - $\delta^* : Q \times \Sigma^* \to 2^Q$.

For every $q \in Q$, $\delta^*(q, \varepsilon) = \varepsilon$-closure$(q)$.

For every $q \in Q$ and $y \in \Sigma^*$, $\sigma \in \Sigma$:

$$\delta^*(q, y\sigma) = \varepsilon\text{-closure}\left(\bigcup \delta(p, \sigma) \mid p \in \delta^*(q, y)\right)$$

A string $x \in \Sigma^*$ is accepted by M if $\delta^*(q_0, x) \cap A \neq \phi$ language L(M) is thus accepted by M. (set of all such strings).



Q2. ε-NFA: $M = (Q, \Sigma, q_0, A, \delta)$.

DFA: $M_1 = (Q_1, \Sigma, q_1, A_1, \delta_1)$.

$Q_1 = 2^Q$.

Initial state q_1 of Q_1 is $\{q_0\}$.

For every $q \in Q_1$, every $\sigma \in \Sigma$,

$$\delta_1(q, \sigma) = \bigcup \{\delta(p, \sigma) \mid p \in q\}.$$

$$A_1 = \{q \in Q_1 \mid q \cap A \neq \phi\}$$

They accept the same language only if for every $x \in \Sigma^*$,

$$\delta_1^*(q_1, x) = \delta^*(q_0, x).$$

when $x = \varepsilon$, $\delta_1^*(q_1, x) = \delta_1^*(q_1, \varepsilon) = q_1 = \{q_0\}$
$= \delta^*(q_0, \varepsilon) = \delta^*(q_0, x)$.

Base case of induction satisfied.

Assume for x, $\delta_1^*(q_1, x) = \delta^*(q_0 x)$. For every $\sigma \in \Sigma$,

To prove: $\delta_1^*(q_1, x\sigma) = \delta^*(q_0, x\sigma)$.

$$\begin{aligned}
\delta_1^*(q_1, x\sigma) &= \delta_1(\delta_1^*(q_1, x), \sigma) \\
&= \delta_1(\delta^*(q_0, x), \sigma) \\
&= \bigcup \{\delta(p, \sigma) \mid p \in \delta^*(q_0, x)\} \\
&= \delta^*(q_0, x\sigma).
\end{aligned}$$

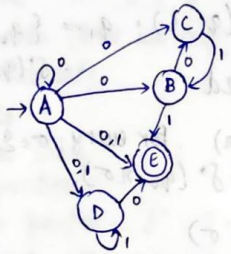
$\therefore$ x is accepted by M_1 only if it is accepted by M.

	ε-closure	0	ε-closure
A	A, B, D	A, C, E	A, B, C, D, E
B	B	C	C
C	C	ϕ	ϕ
D	D	E	E
E	E	ϕ	ϕ

	ε-closure	1	ε-closure
A	A, B, D	D, E	D, E
B	B	E	E
C	C	B	B
D	D	D	D
E	E	ϕ	ϕ

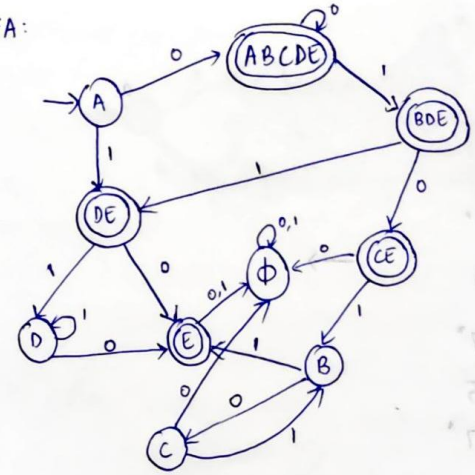
	0	1
→ A	A, B, C, D, E	D, E
B	C	E
C	ϕ	B
D	E	D
* E	ϕ	ϕ

NFA:



	0	1
→ A	ABCDE	DE
* ABCDE	ABCDE	BDE
* BDE	CE	DE
* CE	ϕ	B
* DE	E	D
B	C	E
* E	ϕ	ϕ
C	ϕ	B
D	E	D
ϕ	ϕ	ϕ

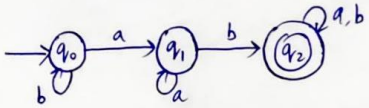
DFA:



Q3. DFA:

a)

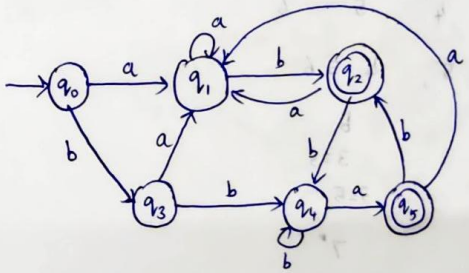
i)



Q3.

a)

ii)



Q3.

b)

	a	b
→ 1	2	3
2	4*	5
3	6*	7*
4*	4*	5
5	6*	7*
6*	4*	5
7*	6*	7*

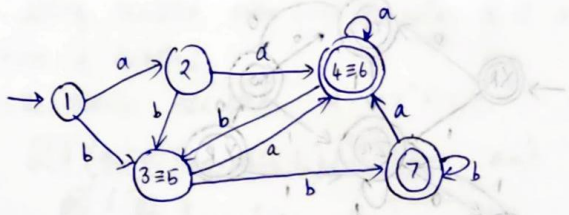
2	×				
3	×	×			
4*	×	×	×		
5	×	×		×	
6*	×	×	×		×
7*	×	×	×	×	×	×
	1	2	3	4*	5	6*

$\therefore$ $3 \equiv 5$, $4 \equiv 6$.

Table:

	a	b
→ 1	2	$3 \equiv 5$
2	$4^* \equiv 6^*$	$3 \equiv 5$
$3 \equiv 5$	$4^* \equiv 6^*$	7*
$4^* \equiv 6^*$	$4^* \equiv 6^*$	$3 \equiv 5$
7*	$4^* \equiv 6^*$	7*

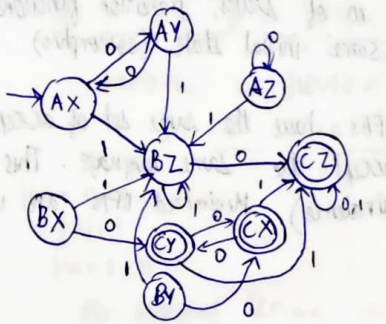
TD:



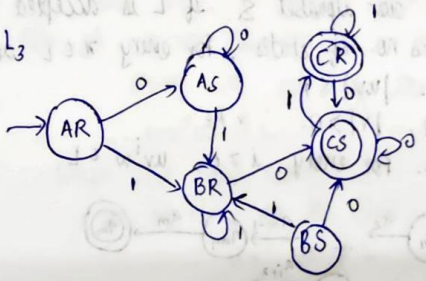
Q4.

a)

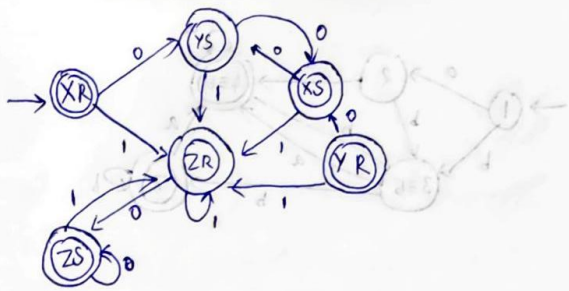
a) $L_1 - L_2$



b) $L_1 \cap L_3$



c) $L_2 \cup L_3$

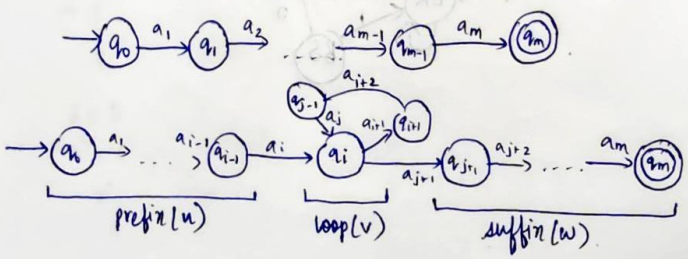


Q4. b) i) Both minimized DFA's of the DFA M and regular expression converted should have same no. of states, transition functions, same set of accepting states and same initial state. (isomorphic).

ii) If it's equivalent minimized DFAs have the same set of accepting states for given values, then they accept the same language. This is an NP-complete decision problem. (equivalence). Minimized DFAs are isomorphic.

Q5.

a) Suppose L is a language over alphabet Σ. If L is accepted by machine $M = (Q, \Sigma, q_0, A, \delta)$, n is no. of states, for every $x \in L$ and $|x| \geq n$, $x = uvw$ such that:
1. $|uv| \leq n$.
2. $|v| > 0$. $v \neq \varepsilon$.
3. For every $i \geq 0$, $uv^iw \in L$.



Minimum string accepted when $i = 0$ is $uv^0w = uw$ as it reaches q_m.

$i = 1$, uvw is accepted.

when $i = 2$, uvvw accepted as $uv^iw \in L$.

$\delta(\delta(q_0, a_1 a_2 \ldots a_{i-1} a_i), a_{j+1}, a_{j+2} \ldots a_m)$

$\delta(\delta(q_i, a_{j+1}, a_{j+2} \ldots a_m))$.

$\delta(q_k, a_{k+1}, \ldots a_m)$.

q_m. Thus, uv^iw is accepted. Hence, proved.

ww or ww^r is not regular.

$$x = \overbrace{\underbrace{1\ldots1}_{n}\ \underbrace{0\ldots0}_{n}}^{w}\ \overbrace{\underbrace{0\ldots0}_{n}\ \underbrace{1\ldots1}_{n}}^{w^r}$$

$|x| \geq n$, let $|uv| \leq n$ $|v| \geq 1$.

$$x = \underbrace{1\ldots}_{u}\ \underbrace{1}_{v}\ \underbrace{0\ldots0\,0\ldots01\ldots1}_{w}$$

$|u| = n - 1$.

$|v| = 1$.

$|uv| = n$.

For pumping lemma $uv^iw \in L$.

when $i = 0$, v doesn't appear.

$\therefore$ No. of zeros on left is 1 less than on the right

$\therefore$ string is not of the form ww^r.

$\therefore uv^0w \notin L$

When $i = 2$ and more,

no. of 1's on the left increases more than right

$\therefore uv^{i \geq 2}w \notin L$.

Similarly for ww:

Q5.

a) ~~No~~.

$$x = \overbrace{\underbrace{1\ldots1}_{n}\underbrace{0\ldots0}_{n}}^{w}\ \overbrace{\underbrace{1\ldots1}_{n}\underbrace{0\ldots0}_{n}}^{w}$$

$$x = \underbrace{1\ldots1\,0\ldots}_{u}\ \underbrace{0}_{v}\ \underbrace{1\ldots1\,0\ldots0}_{w} \rightarrow \underbrace{1\ldots1}_{u}\underbrace{0}_{v}\underbrace{0\ldots0\,1\ldots10\ldots0}_{w}$$

$|uv| = n$.

$|v| \geq 1$

when $i=0$, v doesn't appear

1's on left less than right.

$ww \notin L$.

when $i \geq 2$, 1's on left increases.

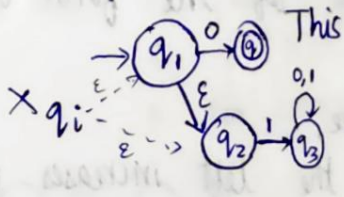
$ww \notin L$.

$\therefore$ ww and $ww^r \notin L$

The language is not regular.

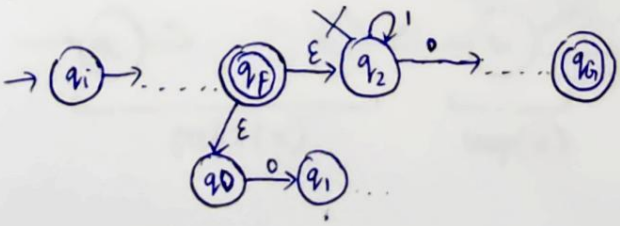
Q5.

b) a) No.



This doesn't work as ε-transition is only from q_i to q_1 and q_2 respectively. There is no ε-transition from q_1 to q_2.

b) No.



If we merge the two states, the ε-closure of the states will change and may accept wrong strings.